ABISHEK GOPAL

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End-to-End Device Validation | Systems Integration | Test Automation

SUMMARY

Systems QA engineer with nine years of experience across consumer hardware and ML-based software, including smart glasses and XR validation at Meta and Emerge. Develops reproducible test procedures and reliable device environments across firmware, connectivity, companion apps, and cloud-backed experiences. Combines hands-on failure diagnosis with Python/TypeScript automation and cross-functional release validation.

EXPERIENCE

META | Burlingame, CA — Aug 2024 – Present
Senior QA Engineer, Wearables - Device, Connectivity & Audio Quality

- Own end-to-end quality for consumer smart glasses across firmware, Bluetooth, iOS/Android companion apps, and cloud-backed assistant services. Validate voice interactions in real-world noise and error recovery across music, calls, messaging, and capture.
- Developed test plans from technical specifications covering user journeys, edge cases, degraded connectivity, and failure scenarios. Led reviews across 4 teams and defined release quality gates and go/no-go criteria for 6 product milestones.
- Triaged 120+ customer-impacting Bluetooth and audio defects and diagnosed 6 critical failure categories using device logs, bug reports, and Bluetooth snoop traces. Isolated host-side versus on-device failures with Audio and Firmware engineering.
- Reproduced a customer-reported audio-routing SEV in a car environment. Compared CarPlay and Android Auto on identical device firmware to isolate host-side behavior, partnered with Audio engineering on the fix, and added multi-device arbitration scenarios to the regression suite.
- Built device-control infrastructure for audio, sensors, and connectivity, reducing firmware-dependent test flakiness by 72%. Added preconditions, topology checks, and recovery procedures that resolved 158 environment failures across 3 test configurations.
- Developed a TypeScript end-to-end automation framework for 7 wearable device families and Android/iOS companion-app pairing. Runs roughly 500 scenarios daily across device and firmware combinations, reporting pass rates, new failures, and suspected failure layers.
- Patched framework APIs and batch-repaired 738 test scripts, restoring the suite to reliable unattended execution.
- Built an AI-assisted pipeline converting TestRail cases into iOS and Android automation scripts with per-step assertions and app-specific media validation.
- Built a TestRail case-quality tool adopted by 3 QA teams, improving test-case completeness from 61% to 89%.
- Co-authored a device architecture reference adopted by 15+ engineers and mentored 3 QA engineers in hardware debugging and ADB diagnostics.

EMERGE.IO (Series B Startup) | Marina Del Rey, CA — Mar 2022 – Aug 2024
QA Lead, Wave-1 XR Consumer Hardware

- Built and led a 5-person QA function, establishing test processes, release criteria, and defect-triage workflows across engineering.
- Authored and maintained 400+ cases covering gesture recognition, haptic accuracy, and rendering. Presented release-readiness assessments and quality metrics to the VP of Engineering at each milestone.
- Isolated a critical gesture-processing memory leak during a 48-hour stress test to unreleased frame buffers in continuous tracking mode, 2 weeks before an investor demo.

- Validated 8 firmware releases across 3 SDK versions, catching 23 compatibility regressions and 4 breaking API changes before integration.
- Built a Python device-orchestration tool for parallel log capture across Meta Quest and Apple Vision Pro, reducing regression environment setup from 2 hours to 20 minutes.

META | Reality Labs | Menlo Park, CA Jun 2020 – Mar 2022
QA Engineer, Imaging & XR

- Validated Ray-Ban Stories camera and imaging behavior across 50+ lighting, skin-tone, and motion scenarios. Traced a 200 ms+ HDR shutter-lag defect to buffer allocation; the fix shipped in pre-launch firmware.
- Ran thermal and battery stress tests under controlled conditions, including 30+ minutes of continuous capture, driving 3 firmware power-management changes. Built an ADB video-logging workflow with sensor overlays that became the standard defect-reporting template.

TESLA | Palo Alto, CA Jan 2020 – Jun 2020
QA Engineer, Autopilot

- Validated computer-vision models across 200+ annotated driving scenarios. Identified a systematic labeling error affecting 15% of edge cases and delivered reports used to prioritize ML retraining.

GOOGLE (via E.Team) | Mountain View, CA Apr 2017 – Dec 2019
QA Analyst, Trust & Safety

- Tested phishing and content-safety classifiers in Chrome and Safe Browsing. Built a QA framework tracking false-positive and false-negative rates across 50K+ daily classification events.
- Led quality reviews across 12 locales, defined precision/recall acceptance thresholds with the ML team, and coordinated 4 offshore QA analysts.

TECHNICAL SKILLS

Connectivity: Bluetooth Classic/LE, A2DP, HFP, AVRCP, LE Audio; pairing, reconnection, audio routing, and multi-device interoperability.

Validation: Structured and exploratory hardware testing; release qualification; regression suites built from field issues; stress, thermal, and battery testing.

Environments and Diagnostics: Device fleets, paired accessories, account/configuration setup, preconditions, and recovery; ADB, device logs, telemetry, and Bluetooth snoop traces.

Automation and Tools: Device orchestration, parallel log capture, Jenkins CI, Android SDK, Xcode, TestRail, JIRA, Git, Postman; AI-assisted test generation.

Languages: Python, TypeScript, Swift, Shell/Bash, SQL.

Platforms and Devices: iOS, Android, Linux, smart glasses, Meta Quest, Apple Vision Pro.

EDUCATION

Doctorate, Applied Computer Science, Westcliff University (in progress)
B.Tech, Information Technology, Bharath University, 2011
MBA, Westcliff University, 2022 **MFA, Animation & VFX**, Academy of Art University, 2016